MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1101- PRECALCULUS

END OF SEMESTER EXAMINATION

Date: 27 August, 2018 Time Allowed: 14:00-17:00)

INSTRUCTIONS

- Write your registration number and level in the space provided below
- Check that the paper has **10** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working

Cohort: _____

Registration Number: _____

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	11	8	3	13	6	4	5	8	8	14	80
Score:											

1. (a) Factorise $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}}$. [4]

(b) Simplify $\dfrac{\frac{4}{a} - a}{1 + \frac{2}{a}}$. [4]

(c) Simplify $\sqrt{2} - \sqrt{12} + \sqrt{48}$ [3]

2. Evaluate

(a) $\sec(-765^\circ)$ [4]

(b) $\csc \frac{4\pi}{3}$. [4]

3. Write as a rational exponent [3]

$$\sqrt{\sqrt[3]{x}\sqrt[4]{y}}$$

4. (a) Let $\log_b 2 = A, \log_b 3 = B$ and $\log_b 5 = C$. Evaluate $\log_b 0.6$ in terms of A, B and C. [4]

(b) Solve the equation $3x(10^x) + 10^x = 0$. [3]

(c) Sketch the graph $y = -\ln(x-2)$, and specify the domain , range, asymptote and intercept. [6]

5. Let $f(x) = \frac{x-1}{3x+5}$.

(a) Find $f(m-2)$. [2]

(b) Find $f^{-1}(x)$. [4]

6. Let $g(x) = \frac{1}{x}$. Find $\dfrac{g(x+h) - g(x)}{h}$. [4]

7. Determine the center and the radius for the circle $x^2 + y^2 + 8x - 6y = -24$. [5]

8. Rationalise the denominator

   (a) $\dfrac{\sqrt{2}}{1-\sqrt{2}}$ [3]

   (b) $\dfrac{1}{\sqrt[3]{a}-1}$ [5]

9. Solve the inequality

$$\frac{x+1}{x+2} > \frac{x-3}{x+4}.$$

[8]

10. (a) Prove the identity $(\sec\alpha - \tan\alpha)^2 = \frac{1-\sin\alpha}{1+\sin\alpha}$.

[5]

(b) Let $y = 2\sin 2\theta$.

i. Find amplitude and period for y. [3]

ii. Sketch the graph of y. [6]